

Partial Result: TC/KR Density for Tight Measures on Arbitrary Metric Spaces

Automatic Math Discovery Run `fa.banach.001`

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Source Problem

The source paper is Sofiya Ostrovska and Mikhail I. Ostrovskii, *Generalized transportation cost spaces*, arXiv:1902.10334 [1]. Theorem 1.13 proves that if M is Polish, then $(\text{TP}(M), \|\cdot\|_{\text{TC}})$ is dense in the Kantorovich-Rubinstein measure space $(\mathcal{M}, \|\cdot\|_{\text{KR}})$. Remark 1.14 says that the authors do not know how to prove an analogue for general metric spaces. In the parsed source this is `data/parsed/arxiv_sources/1902.10334/source.tex`, lines 680–746, and in the PDF it is page 11.

Next, we split K into a finite number $\{K_n\}_{n \in T}$ of pairwise disjoint Borel subsets of diameter $< \varepsilon/(3(\mu(M) + \nu(M)))$ each. Define a function f on M as follows: In each of the sets K_n , we pick a point t_n and let $f(t_n) = \mu(K_n) - \nu(K_n)$ for $n \in T$. If f is extended as 0 to the rest of M , the obtained function is not necessarily in $\text{TP}(M)$. To balance this, the 0-extension is modified at point O by taking $f(O) = -\sum_{n \in T} f(t_n)$, thus implying $f \in \text{TP}(M)$. If $t_n \neq O$, the following inequality holds:

$$\begin{aligned} \left| \int_{K_n} l(v) d(\mu - \nu)(v) - l(t_n) f(t_n) \right| &= \left| \int_{K_n} (l(v) - l(t_n)) d(\mu - \nu)(v) \right| \\ &\leq \int_{K_n} |l(v) - l(t_n)| d\mu(v) + \int_{K_n} |l(v) - l(t_n)| d\nu(v) \leq \frac{\varepsilon}{3} \cdot \frac{\mu(K_n) + \nu(K_n)}{\mu(K) + \nu(K)}, \end{aligned}$$

where, in the last inequality, the fact that l is 1-Lipschitz and the assumption on the diameter of K_n are used.

Since $l(O) = 0$, one has:

$$\begin{aligned} |l(\mu - \nu) - l(f)| &\leq \sum_{n \in T} \left| \int_{K_n} l(v) d(\mu - \nu)(v) - l(t_n) f(t_n) \right| + \left| \int_{B(O, R) \setminus K} l(v) d\mu(v) \right| \\ &\quad + \left| \int_{B(O, R) \setminus K} l(v) d\nu(v) \right| + \left| \int_{M \setminus B(O, R)} l(v) d\mu(v) \right| + \left| \int_{M \setminus B(O, R)} l(v) d\nu(v) \right| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} = \varepsilon. \end{aligned}$$

□

Remark 1.14. We do not know how to prove an analogue of Theorem [1.13](#) in the case of general metric spaces. See [\[Bil68\]](#) p. 234–235 in this connection. Nonseparability will not be an obstacle if we consider uniformly discrete spaces, because in such spaces finite Borel measures have countable support. The general metric case is beyond the subject of this work. For a systematic exposition of generalizations of results of Kantorovich and Rubinstein we refer the reader to [\[RR98\]](#).

Finally, it is worth mentioning that in Computer Science the transportation cost is often called *Earth Mover's Distance*. This name was introduced by Rubner-Tomasi-Guibas [\[RTG98\]](#) in their work on computer vision. They knew the notion of transportation cost and their goal was to generalize it to the cases where total demand can be less than the total supply. This more general case is not considered in the present paper as such generalized transportation problems do not form a vector space.

Several authors published their opinions on the most suitable choice of the name for the spaces which we call *transportation cost spaces*. Vershik [\[Ver04\]](#) [\[Ver13\]](#) provided an argument in favor of the name *Kantorovich space*. Villani [\[Vil09\]](#) pp. 106–107] decided in favor of Wasserstein, and Weaver [\[Wea18\]](#) p. 125] defended the name *Arens-Eells space*. It is interesting to mention that Villani decided in favor of

Idea of the Proof

The Polish proof uses separability and completeness only through a compact approximation step: Ulam's theorem produces a compact set capturing almost all of the mass inside a large ball. If the measures are assumed tight, that compact set is available in any metric space. The rest of

the source proof is purely metric: partition the compact set into finitely many small Borel pieces, replace the measure on each piece by one atom, and use the Lipschitz bound to control all test functions simultaneously.

Definitions

Fix a pointed metric space (M, d, O) . Let $\text{TP}(M)$ be the vector space of finitely supported real functions f on M with $\sum_x f(x) = 0$. For a finite signed measure σ with total mass zero and finite first moment, write

$$\|\sigma\|_{\text{Lip}} = \sup \left\{ \left| \int_M l d\sigma \right| : l(O) = 0, \text{Lip}(l) \leq 1 \right\}.$$

For finitely supported $\sigma = f$, this is the transportation-cost norm.

Tight-Measure Analogue

Partial Theorem. *Let (M, d, O) be an arbitrary metric space. Let μ, ν be tight finite positive Borel measures on M , with equal total mass and finite first moment. Then for every $\varepsilon > 0$ there is $f \in \text{TP}(M)$ such that*

$$\|\mu - \nu - f\|_{\text{Lip}} < \varepsilon.$$

Consequently, the finite transportation problems are dense in the tight finite-measure part of the Kantorovich-Rubinstein space. Equivalently, whenever the KR transport norm is identified with the Lipschitz-dual norm on this measure class, the density conclusion of Theorem 1.13 remains valid without assuming M is Polish.

Proof. Put $\lambda = \mu + \nu$. If $\lambda(M) = 0$, take $f = 0$, so suppose $\lambda(M) > 0$. Let $\varepsilon > 0$. Since μ and ν have finite first moment, choose $R > 1$ such that

$$\int_{M \setminus B(O, R)} d(O, x) d\lambda(x) < \varepsilon/3.$$

Since λ is tight, the restriction of λ to the closed ball $B(O, R)$ is tight. Hence there is a compact set $K \subset B(O, R)$ such that

$$\lambda(B(O, R) \setminus K) < \frac{\varepsilon}{3R}.$$

Partition the compact metric space K into finitely many Borel sets $\{K_j\}_{j=1}^N$ of diameter less than $\varepsilon/(3\lambda(M))$. Choose $t_j \in K_j$, put $a_j = \mu(K_j) - \nu(K_j)$, and define f by

$$f = \sum_{j=1}^N a_j (\delta_{t_j} - \delta_O).$$

Then $f \in \text{TP}(M)$, even if one of the selected points t_j is O .

Let l be any 1-Lipschitz function on M with $l(O) = 0$. On each piece K_j ,

$$\left| \int_{K_j} l d(\mu - \nu) - l(t_j) a_j \right| \leq \int_{K_j} |l(x) - l(t_j)| d\lambda(x) \leq \frac{\varepsilon}{3\lambda(M)} \lambda(K_j).$$

Summing over j gives a total contribution at most $\varepsilon/3$. On $B(O, R) \setminus K$, the bound $|l(x)| \leq R$ gives

$$\left| \int_{B(O, R) \setminus K} l d(\mu - \nu) \right| \leq R \lambda(B(O, R) \setminus K) < \varepsilon/3.$$

On $M \setminus B(O, R)$, the bound $|l(x)| \leq d(O, x)$ gives

$$\left| \int_{M \setminus B(O, R)} l d(\mu - \nu) \right| \leq \int_{M \setminus B(O, R)} d(O, x) d\lambda(x) < \varepsilon/3.$$

Combining the three estimates yields

$$\left| \int_M l d(\mu - \nu) - \int_M l df \right| < \varepsilon$$

uniformly over all such l . This is exactly $\|\mu - \nu - f\|_{\text{Lip}} < \varepsilon$. $\square$

KR Norm for Tight Measures

Lemma. *For tight finite Borel measures μ, ν as above, the usual W_1 or Kantorovich-Rubinstein transport norm equals the Lipschitz-dual norm $\|\mu - \nu\|_{\text{Lip}}$.*

Proof. Let $\lambda = \mu + \nu$. Tightness gives compact sets $C_n \subset M$ with $\lambda(M \setminus C_n) < 2^{-n}$. Replacing C_n by $\bigcup_{k \leq n} C_k$, we may assume C_n is increasing. The set

$$S = \bigcup_{n=1}^{\infty} C_n$$

is sigma-compact and separable, and $\lambda(M \setminus S) = 0$. Thus both measures are supported on the Borel subspace S .

The completion $\widehat{S}$ is Polish. The measures extend to finite Borel measures on $\widehat{S}$ supported on S , with the same first moments. The Polish Kantorovich-Rubinstein duality applies on $\widehat{S}$. Couplings of the extended measures are supported on $S \times S$, so the transport infimum computed in M , in S , or in $\widehat{S}$ is the same. Finally, every 1-Lipschitz function on S extends to M , and also to $\widehat{S}$, by the McShane extension formula. Hence the Lipschitz-dual suprema are the same in all three spaces. This proves the claimed identification. $\square$

Scope and Limitations

This is a partial result. It proves the general-metric analogue of Theorem 1.13 for tight finite Borel measures, including finite Radon measures. It does not settle the full generality hinted at in Remark 1.14: arbitrary finite Borel measures on arbitrary metric spaces may fail to be tight, and the compact approximation step above gives no information in that case.

Novelty Search Bounds

Before promotion, the run registry, solution index, attempt index, and proof-gap index were searched for this arXiv identifier and for the keywords “generalized transportation cost spaces”,

“TC/KR”, “tight measures”, and “general metric spaces”. No existing packet for this source remark was found. Web searches on June 16, 2026 for close phrases around “Kantorovich-Rubinstein tight measures arbitrary metric space” found standard KR-duality material, but no exact later answer to Remark 1.14 in this run’s packet sense.

Human Review Recommendation

Review the separable-support reduction in the KR-norm lemma and the replacement of Ulam’s theorem by tightness in the density proof. If those two points are accepted, the partial theorem is a direct extension of the source proof.

References

- [1] S. Ostrovska and M. I. Ostrovskii, *Generalized transportation cost spaces*, arXiv:1902.10334, 2019.